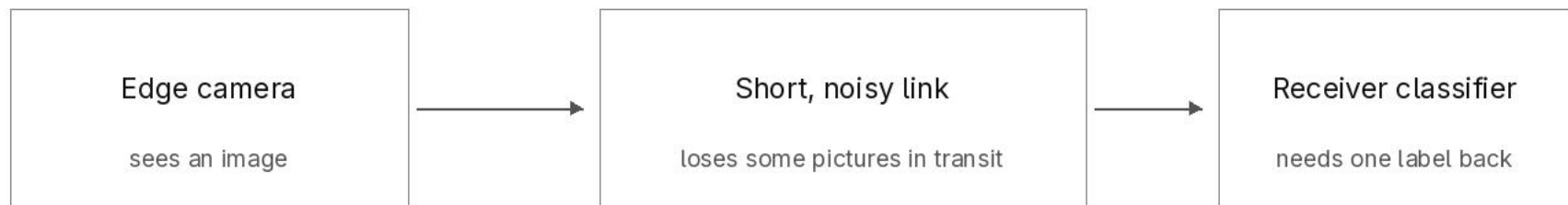


Task-Oriented Deep Joint Source–Channel Coding

What a camera should send when the receiver only needs the answer



Train the sender and the receiver together so the classification decision survives a short, noisy link.

The thesis

Second Review, 29 September – 3 October 2026.

What has changed since the first review

In August the digital system was working. The learned models are trained now.

Aug · Digital system	Measured how badly it loses pictures at every setting, tested it on all 288,000 held-back combinations, and froze the settings it should use at each signal strength.
Aug · Learned system	Built the training loop, trained models at several seeds, and froze three of them at half rate.
Sep · Fair comparison	Built a control that sends learned features digitally, so the comparison includes a task-aware system alongside a compressed image.
Sep · Weak-link training	Trained a second version with the signal strength picked at random, and compared it against the first.
Sep · Crossing question	Decided it on held-back pictures before seeing the answer: the two lines cross between -5 and -4 dB.
Sep · Power limits	Trained one model under a hard limit on how tall the signal spikes may get.
Oct · Measurements	252 measurements across twelve versions of the system and twenty-one signal strengths, 1,000 held-back pictures each.
Nov · Exam set	Still locked. It opens in week 11 once the locking document is signed.

Everything above ran on held-back pictures. The exam set stays locked until gate G-12.

Objectives, stated as completion criteria

Five things we set out to build. Each one is checkable.

What we set out to do

- | | |
|----------------------------|--|
| 1. Build it | Train a learned sender and receiver together, so the picture goes through the same kind of noise a radio adds. |
| 2. Same radio time | At every signal strength and rate, both systems get the same number of chances to use the radio. |
| 3. Charge the overhead | Checksums, padding and error-correction bytes come out of the budget before any picture data. |
| 4. Pick the setting fairly | The operating point is chosen using the digital system alone, before anyone sees the learned curve. |
| 5. Report per picture | Every row records what happened to one picture, including those that failed to arrive. |

How these are scored

The rubric scores whether this list was completed.

Completion is independent of which way the curve went. A crossing and the learned system winning everywhere both finish the list on the left.

Before the numbers, one picture

Every chart that follows runs from a very weak radio signal on the left to a clean one on the right.

Plain words	Our term	Where you see it
How strong the signal is	SNR, in decibels	the left-to-right axis
How many pictures got through	delivery coverage	the flat 10% line
How many we labelled correctly	top-1 accuracy	the up-and-down axis
How many times we use the radio	channel-use budget	12,800 or 3,200 per picture

Every point on every chart is the same 1,000 pictures, at that one signal strength.

One example

At the weakest signal we measured:

The learned system labelled 728 of the 1,000 pictures correctly.

The digital system got almost nothing through, so every picture fell back to one fixed answer — and that answer happens to be right 100 times out of 1,000. So 10%.

That flat 10% is the fallback rule: 1,000 pictures that never arrived.

What we measured for this review

We built twelve versions of the system and tested each one at twenty-one signal strengths.

Item	Figure	What it is
Measurements	252	One row each in the published record
Versions of the system	12	Five of them are controls rather than headline systems
Signal strengths	21	From very weak to very clean
Pictures per point	1,000	The same held-back pictures every time
Exam pictures opened	0	Sealed until gate G-12

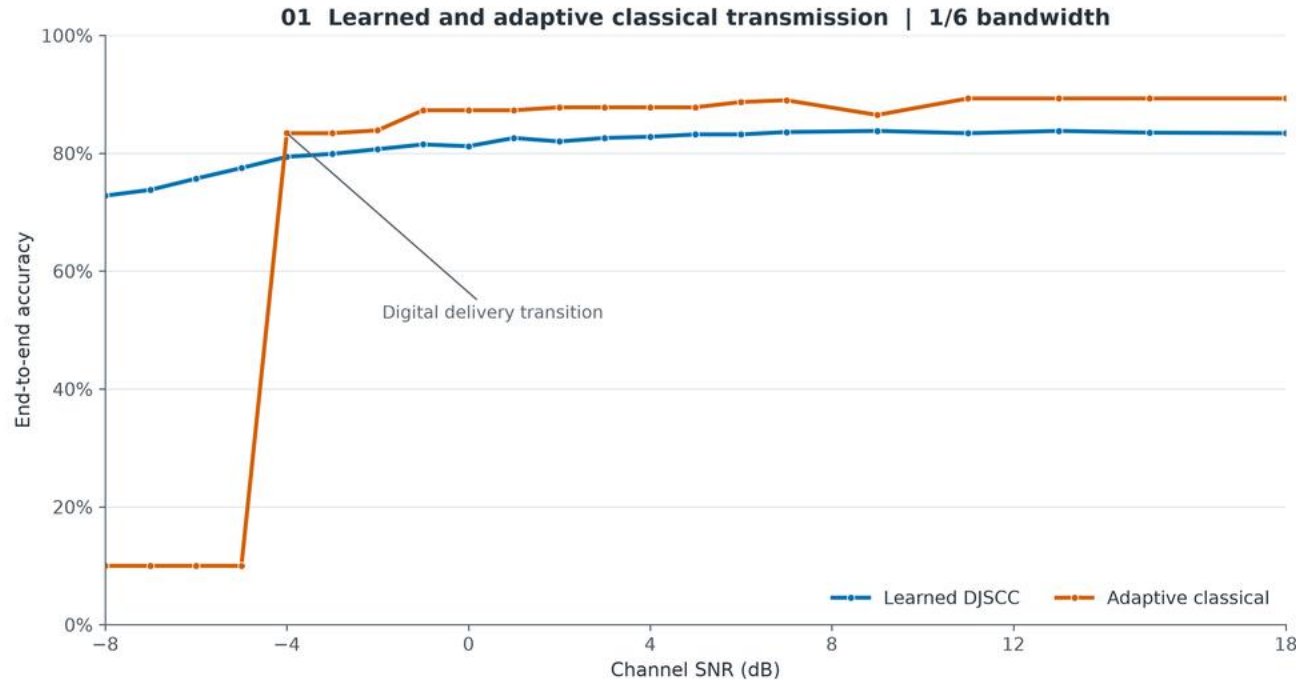
Two things that govern how every later chart should be read

Radio time Half rate uses the radio 12,800 times per picture; quarter rate uses it 3,200 times. Within one rate both systems get exactly the same amount of radio time.

Who marks it The learned systems are marked by the classifier built into them. The digital systems are marked by one we trained on compressed-looking images. So we compare whole systems: the link, the representation, and the marker.

Where the digital system gives up, and where it takes over

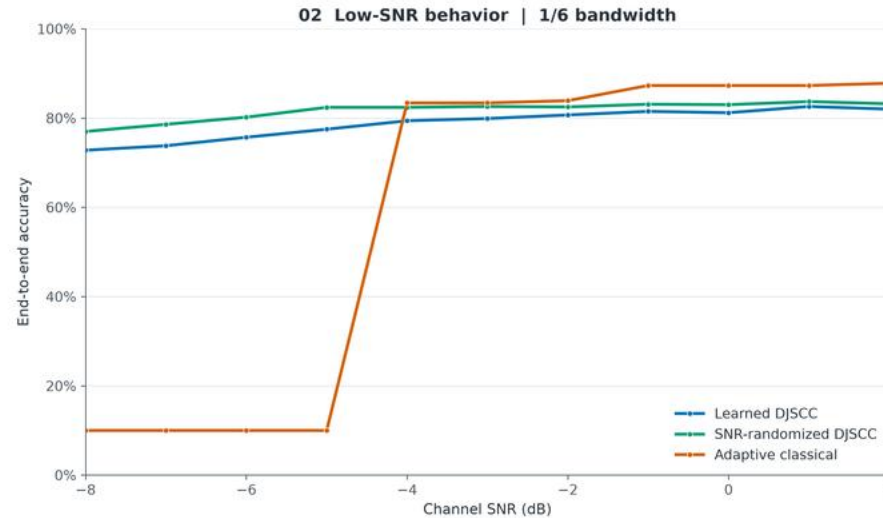
The learned system keeps answering the whole way across. The digital system stays silent, then overtakes it.



At the weakest signal the learned system is right on 728 of 1,000 while the digital system delivers nothing. From -4 dB it delivers and goes ahead. At +18 dB it is well ahead.

Close-up: the moment the digital system starts working

Between -5 and -4 dB the digital system begins getting pictures through, and its score jumps 73 points.



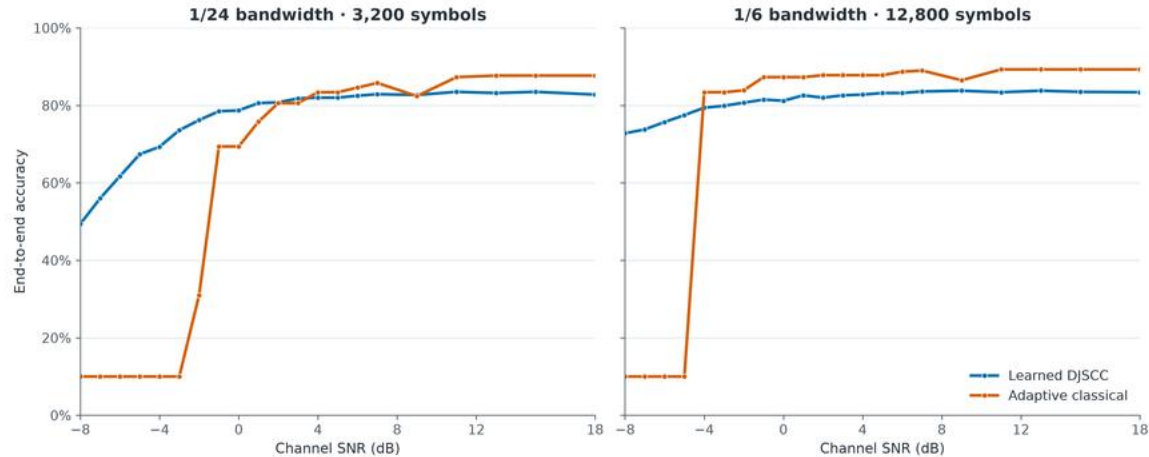
At -8 dB The learned system is right on 728 of 1,000. The digital system got nothing through, so it scores 10%.

At -4 dB The digital system now gets everything through and reaches 83.4%, ahead of the learned system's 79.4%.

After that It dips at -2 dB and again at $+9$ dB.

What happens when we quarter the radio time

On a quarter of the radio time, the digital system loses far more at the weak end than the learned one does.



Weak signal

Going from half to quarter rate drops the learned system from 72.8% to 49.4%. That is a 23 point loss.

At 0 dB, quarter rate

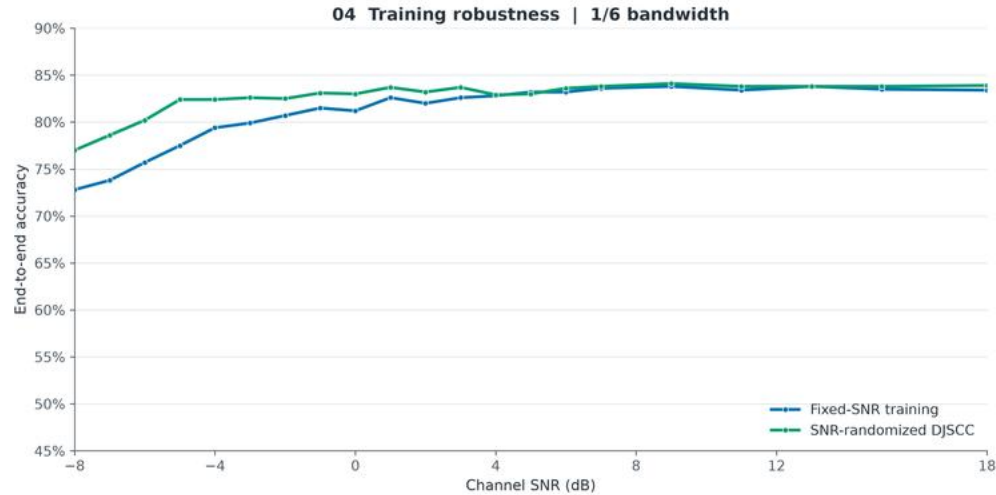
Learned 78.7% against digital 69.4%. The learned system is still ahead.

Catches up, then slips

Digital passes learned at the +4 dB point, loses the lead at +9 dB, then goes ahead again.

Training across signal strengths instead of one

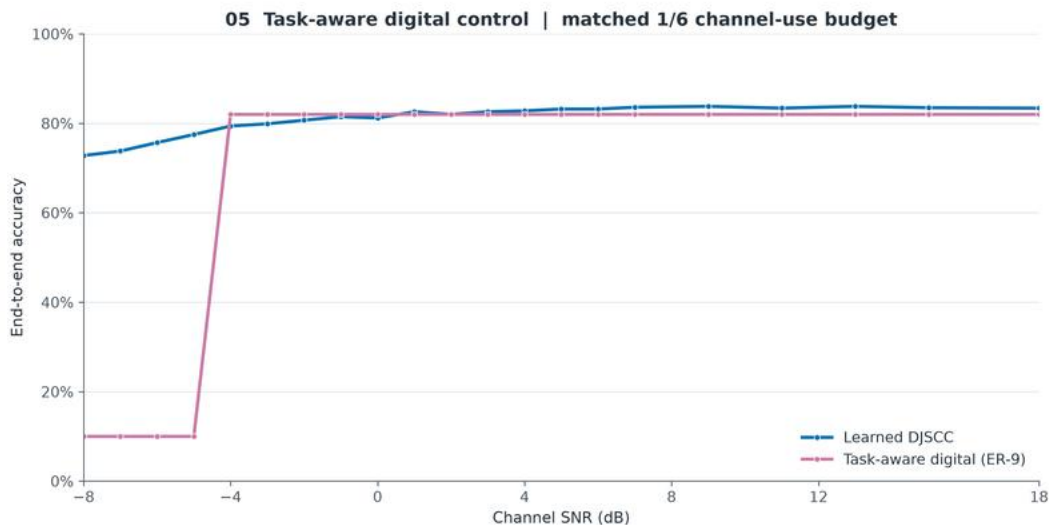
Picking the training signal strength at random each time lifts the whole weak end of the line.



Training	-8 dB	-4 dB	0 dB	+9 dB	+18 dB
Random each time	77.0	82.4	83.0	84.1	83.9
One fixed strength	72.8	79.4	81.2	83.8	83.4

The control that keeps the comparison honest

If we had only compared against a normal compressed image, the weak-signal story would be too easy to believe.



What this arm does

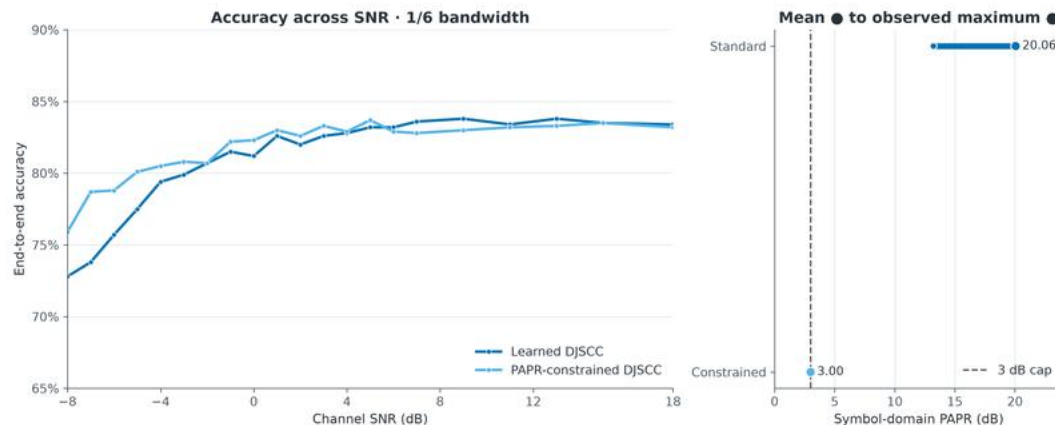
Instead of a JPEG image it sends the learned features themselves, through the same error-correction code and radio settings, using the same 12,800 uses of the radio. It separates "knows the task" from "codes both at once".

What it settles

It holds 82.0% at every strength from -4 dB up. Sending learned features digitally is strong once pictures get through, so the weak-signal gap is about failed deliveries as much as representation.

Keeping the signal peaks from getting too tall

A real transmitter needs a limit on signal peaks, so we trained a second model under one.

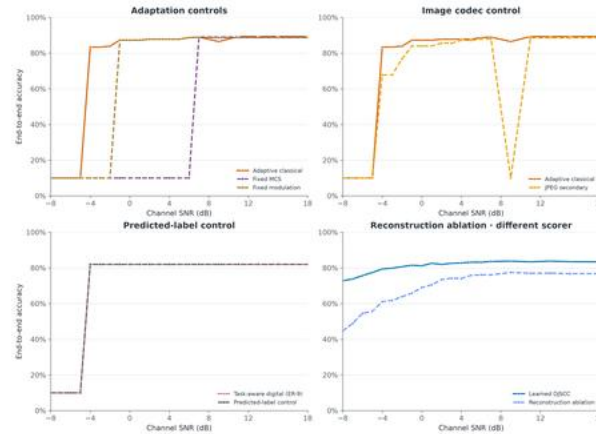


“Peak-to-average power ratio” = how much taller the tallest spike is than the average.

Model	Tallest spike	At -8 dB	At +18 dB
Power-limited	3.000002 dB	75.9%	83.2%
Ordinary learned	20.059284 dB	72.8%	83.4%

Four controls, including one that went against us

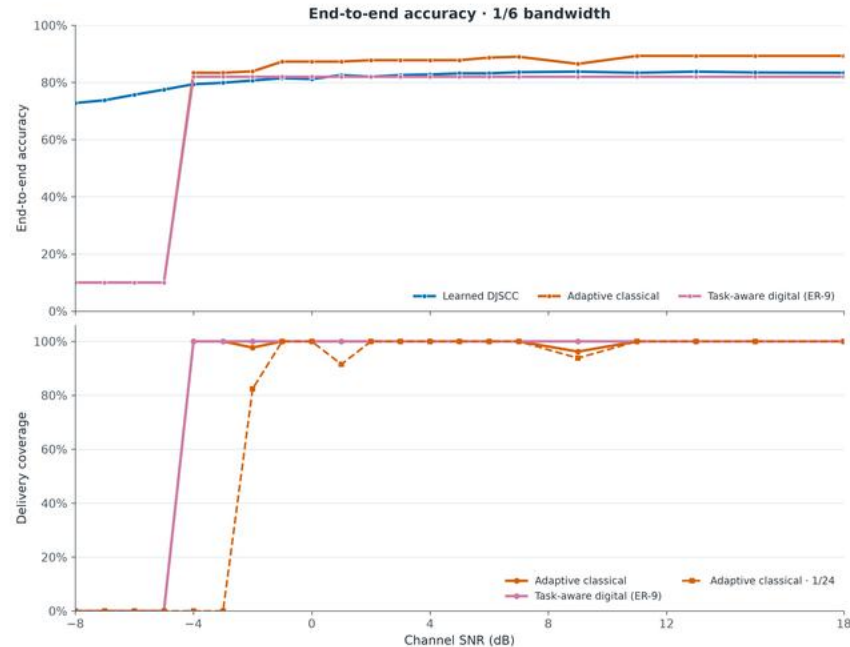
Each panel takes one design choice away from the digital system and shows what it was worth.



Fixed settings	Sits at the fallback floor until +6 dB, then reaches 89.0%. Choosing settings per strength buys real reach.
Fixed modulation	Starts working later than adaptive selection, and settles near 88.7%.
JPEG secondary curve	88.6% at +18 dB, with a full outage at +9 dB.
Sending the answer	82.0% once it arrives — the sender's own predicted label.

Why the digital line breaks where it does

The score only moves when the pictures start or stop arriving.



The 10% is a rule

Every picture takes the same fallback answer when delivery fails. On an evenly split set that is right 100 times in 1,000.

Outages are part of it

We report how many pictures arrived next to how many we got right, so a broken link is distinguishable from a wrong guess.

Learned always answers

Its rules promise a label at every strength, so its curve never flattens against a floor.

The numbers behind the curves

How many of the 1,000 held-back pictures we labelled correctly, one picture at a time.

Half rate — 12,800 uses of the radio

System	-8	-5	-4	0	+9	+18
Learned, one strength	72.8	77.5	79.4	81.2	83.8	83.4
Digital, picks settings	10.0	10.0	83.4	87.3	86.5	89.3
Learned, random	77.0	82.4	82.4	83.0	84.1	83.9
Digital, learned feats	10.0	10.0	82.0	82.0	82.0	82.0
Learned, power-limited	75.9	80.1	80.5	82.3	83.0	83.2

Quarter rate — 3,200 uses of the radio

System	-8	-4	-2	0	+4	+18
Learned, one strength	49.4	69.3	76.2	78.7	82.0	82.8
Digital, picks settings	10.0	10.0	31.0	69.4	83.4	87.7

Where the two lines cross

At half rate, which system is ahead flips between -5 dB and -4 dB. We decided the crossing question on exactly that, before seeing the answer.

Reading the two tables

Each column is one signal strength; each cell is the number of correct labels out of the same 1,000 held-back pictures. At half rate the learned row leads up to -5 dB and the digital row leads from -4 dB. At quarter rate the digital row stays on the fallback floor four dB longer, then climbs past the learned row from +4 dB. The difference between the two tables is the effect of cutting radio time fourfold.

How to read these results

One training run

Every figure is a single observed measurement from one run over 1,000 held-back pictures.

Whole systems

The learned systems are marked by their own classifier, the digital systems by one trained on compressed images. These numbers describe whole systems.

Different amounts of work

The digital settings and grader were frozen from earlier validation work. The learned versions are separate training runs.

Simulated channel

Peak-to-average ratio is measured on the numbers we send. Real-radio replay is scheduled for W14.

Frozen operating points

The fallback rule and fixed settings produce visible jumps. Curves pass through measured points only.

Both outcomes count as success under the specification: a crossing, and the learned system winning everywhere.

Four times we changed the approach because of what we found

The rubric asks what we decided based on the results obtained. These are the four that count.

AM-34

We had pinned the digital system to the weakest radio setting, which made the lines crossing impossible by arithmetic. We let it pick a stronger setting as the signal improves.

The rule we adopted: every change strengthens the digital system or is decided in advance.

AM-52

Our signal strengths were spaced 2 dB apart exactly where the sharpest drop happens, so we could have missed it entirely. We added three more strengths before it mattered.

Put the detail where the effect is.

AM-58

The script that checks our packet sizes reported no problems while breaking four of its own rules. We fixed the script, and every result came out the same.

All 215 configurations stayed feasible, and every scientific outcome was unchanged.

AM-60

The exam set was set to unlock in week 9, about three weeks before the manifest that locks it existed. We moved that to week 11.

They have stayed locked ever since.

Twenty-two further changes are written down in SPEC §17, including three checks that turned out to be wrong and had to be repaired.

What we claim as new

Both claims are about how the experiment is built.

The task-aware digital control

A digital arm sends learned features through the same error-correction code and radio settings, using the same amount of radio time. It tells "knows the task" apart from "codes both at once" inside one experiment.

Prior work reports the gap without this control.

Charging the overhead to the digital system

The picture file, framing, checksums and rate-matching bytes are all measured and counted, so the digital system is tuned on the payload it can genuinely use.

Prior work quotes a nominal rate and leaves the overhead uncounted.

What this builds on

- Getting steadily worse as the link worsens
- Avoiding the sudden drop-off
- Sending task-aware features digitally
- Training a neural encoder and decoder

All four are established in the prior literature. This project builds on them.

Negative check

Each claim resolves to a specific reference in the 30-source review at docs/literature-review.md. The full written novelty statement ships with the final review package.

Where the remaining weeks went

W10	29 Sep – 3 Oct	Second Review	Held-back measurements complete; package and figures frozen	done
W11	5 – 11 Oct	Open the exam set	One guarded opening once the locking manifest is signed	planned
W12	12 – 18 Oct	Decide everything	Every hypothesis settled on the exam set; first tier frozen	planned
W13	19 – 25 Oct	Demo	Signal-strength slider, both systems side by side, frozen plot	planned
W14	26 Oct – 1 Nov	Poster and hardware	Poster draft; real radio only if the first tier leaves room	planned
W15	2 – 8 Nov	Internal freeze	Report finished in the required format	planned
W16	9 – 15 Nov	Spare week	Report completion and checking	planned
W17	16 – 22 Nov	Final Review	Final review 17–21 Nov; report and material due 20 Nov	planned

The report is due inside Final Review week, so W15 is an internal freeze and W16 is reserved for report completion.

Where this leaves the project

The learned system keeps answering when the link is too weak for the digital one to get a single picture through. Once the digital system starts delivering, it is the more accurate of the two. They cross between -5 and -4 dB.

Next One guarded run on the exam set once the locking manifest is signed. So far, zero exam pictures opened.

Then Every hypothesis settled on the exam set, and the first tier frozen.

Later Real-radio replay and the live demo run once the first tier is frozen.

Evidence trail

results/learned/w10/ holds the 252-measurement record and the per-picture manifest. Figures and data are in presentation-results/. Requirements and changes are in spec/SPEC.md §17.

The exam set stays sealed until gate G-12.